scpPower M1 plausibility note − frozen addendum v1.1.1

13 October 2026. Every specified assay was evaluated; both Derks instruments are separate. Values below are unchanged estimates, not targets for tuning. Cultures are cell lines, not donors. Missingness uses observed pre−shift protein means; only variances receive per−cell median shifts.

Specht − FLAG / dataset stopped. Missing 79.5%, mean 624.5 proteins/cell and median peptide−agreement cell CV 0.407. The paper implies about 79.5% missing and 622.7 proteins/cell from 305 cells/protein; its 10.0% peptide missingness is conditional on within−set identification, not global completeness. Coverage is consistent. Its reconstructed CV cannot reproduce the pre−normalisation QC metric exactly (FLAG: processed peptide input). However, the specified peptides is already log2 and row/column normalised (source export cr_norm_log(t3)); raw abundance cannot be recovered. Quantification slope −8.118 [−8.246, −7.991] is implausibly negative. These tier/curve/variance estimates remain diagnostic. Specht.

Leduc − FLAG / dataset stopped. Missing 70.8%; mean 1004.3 proteins/cell versus the paper's 997 (implied missing 64.9%); median peptide−agreement cell CV 0.396 versus the source QC cutoff 0.41 (median below cutoff; source does not state an overall central cell CV). FLAG: coverage differs after explicit reverse/contaminant exclusion and reaggregation, with no filtering added to force agreement. peptides, not peptides_log, was used: the latter changes features, cells and normalisation. Input is reference−channel ratios. Quantification slope −0.776 [−0.780, −0.773]; FLAG: significantly negative; no corrective refit. Leduc.

Derks − timsTOF / Q Exactive. Global missing 33.0% / 59.6%; mean proteins/cell 1015.3 / 474.1 versus approximately 1000. Within−plex completeness 98.8% / 86.1%: timsTOF agrees with the published >98% claim; global and within−plex denominators are distinct. FLAG: Q Exactive coverage is below the headline count; instrument/gradient and early−assay scope differ. Median observed protein CV across cells 0.794 / 0.748; the paper's repeat−injection CV is not cell−to−cell CV, so exact cell−CV agreement is untestable. File/label matching corrected permuted metadata, with verified instrument and single−cell identities. Derks.

Brunner − calibrated missingness, unavailable variance. Raw protein missing 48.1%; mean 1284.2 proteins/cell, minimum 484 versus the paper's 600 proteins/cell cutoff (FLAG: below it). Median observed linear protein CV 0.350 versus reported 0.3: FLAG, different upstream processing/QC. No missingness percentage matching this raw assay is stated in the paper; completeness is abundance−dependent. Each run contains one HeLa cell, so both residual−within−run and separate run variance are unidentifiable; explained NA, not zero. Brunner.

Walzik / compression. 23 healthy donors, 46 baseline observations, exact identifier overlap (Specht, Leduc, timsTOF, Q Exactive, Brunner): 1862, 2579, 781, 603, 1553. Proteomics is whole PBMC; subsets are flow cytometry, not sorted−subset proteomics. Lineage−specific donor variance is unavailable. Whole−PBMC variance includes donor composition differences and remains an unvalidated proxy for these cell lines. Walzik. k=1 is a registered assumption, not estimated; sensitivity assumptions are 0.75 and 0.5. The lab pilot needs a two−proteome ratio standard. The compression plot is dropped. Cheung, Nature Methods.

Reproduction: inst/scripts/calibrate_resumed.R and report_resumed.R; seed 20261013. Software: R=4.6.1; scp=1.22.0; limma=3.68.5; edgeR=4.10.5; scplainer=workflow in scp 1.22.0; no separate package